Deep Learning-Based Method for Predicting Student Dropouts in MOOCs

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***Abstract*—Massive Open Online Courses (MOOCs) have attracted a large number of students due to their unique teaching methods and high-quality course resources. However, the high dropout rates present a significant challenge, affecting the sustainable development of MOOC platforms and diminishing students' learning effectiveness. To address this issue, this paper proposes a deep learning-based model to predict student dropout behavior. The experimental data is sourced from the MOOCCube2020 dataset. Initially, a convolutional neural network (CNN) is used to extract implicit features from temporal feature data, which are then input into a bidirectional long short-term memory (BiLSTM) network. Subsequently, a fusion attention mechanism recalculates(ATT) the hidden layer states output by the BiLSTM layer. The final result is obtained through continuous training and optimization. Experimental results demonstrate that the model effectively captures the correlation between learning behaviors and dropout behaviors, enabling accurate predictions of student dropouts.**

Keywords—MOOC，Dropout prediction，Learning behavior data, Time Series Model

I. INTRODUCTION

Massive Open Online Courses (MOOCs) represent an innovative educational approach that has emerged alongside the rapid development of Internet technology [1]. MOOCs aim to meet the increasing global demand for online education by providing an Internet-based platform that facilitates convenient connections and communication between learners, educators, and institutions. Learners can easily access high-quality educational resources, engage with prestigious institutions, interact with instructors and peers worldwide, and earn course certificates [2]. The advantages of MOOCs, such as being free, convenient, and efficient, have attracted a large number of participants. However, MOOCs face a significant challenge: high dropout rates[3]. Studies indicate that only a small percentage of learners complete their courses. For example, among the 17 MOOCs offered by Harvard University and the Massachusetts Institute of Technology (MIT), only 5.1% of participants earned certificates. This high dropout rate is a major obstacle to the growth of MOOCs, necessitating effective solutions [4]. To investigate the reasons behind MOOC dropout rates, researchers have employed various methodologies, including statistical analyses, interviews, questionnaires, learner profiling, social network analyses, and sentiment detection [5]. Although comprehensive analytical frameworks are still developing, existing studies have proposed valuable methods that lay the groundwork for further research[6]. This study examines the MOOC domain to predict learner dropout behavior by analyzing learner behavior data[7]. The study engineered features from learner behavior data and analyzed student learning behavior data on a weekly basis. Dropout behavior was predicted using a model developed in this study, which combines convolutional neural networks, bidirectional long short-term memory (BiLSTM) recurrent neural networks (RNN), and attention mechanisms (ATT). By analyzing learner behavior data, it is possible to understand learning patterns, identify potential dropout risks, and provide targeted support to enhance learning quality and completion rates.

II. RELATED WORK

A. Definition of dropout

The concept of dropout originates in traditional classrooms, where it refers to students who fail to meet teachers' requirements and prematurely discontinue their studies[8].Compared to traditional learning environments, online learners exhibit more complex and diverse motivations. Therefore, to facilitate research on dropout behavior analysis, this study establishes a specific definition of online learning dropout. Existing approaches to predicting MOOC dropouts vary among researchers. Two common definitions are presented:

- Definition 1: Learners who exhibit no engagement during the final period before the course concludes[9].
- Definition 2: Learners' overall activity throughout the course and their participation in continuous learning activities[10].

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These definitions differ in their focus: Definition 1 emphasizes the final learning state to assess student dropout and is often used as a criterion for predicting dropout. However, in engineering education certification, instructors often prioritize tracking student mastery of course knowledge throughout the learning process, or within specific course segments. Definition 2 aligns more closely with this objective by considering students' engagement at various stages of the learning process. Consequently, this study adopts an approach similar to Definition 2 to define the issue of student dropout in online learning platforms.

B. Relevant Methods

When building a prediction model for online learning dropout, there are three common modeling methods: machine learning, ensemble learning, and deep learning methods[11]. As the number of online learners increases, the complexity of learning behavior data has also grown, characterized by high dimensionality, dynamics, and nonlinear correlations, presenting new challenges to dropout prediction. In the field of dropout prediction, the limitations of traditional machine learning methods are becoming apparent, while deep learning-based approaches are gaining popularity.

Lu et al. extracted various behavioral features, such as clickstreams, homework tests, and forum behaviors, among others, from raw data. They proposed a sliding-window-based MOOC dropout prediction model that can dynamically track students' dropout risk over time[12]. Xia Sun et al. developed a CNN_LSTM model that combines a Convolutional Neural Network (CNN) with a Long Short-Term Memory Network (LSTM). This model extracts abstract and effective data on student behavioral features using CNNs and then dynamically predicts MOOC student dropout through LSTM models[13].Yin et al. proposed a deep learning model incorporating a self-attention mechanism and a conditional random field architecture to extract key features and improve feature representation quality by mapping individual features to a high-dimensional space, followed by reconstruction using the self-attention mechanism and CNNs[14]. Feng et al. introduced a context-aware Feature Interaction Network (CFIN) based on CNNs for MOOC dropout modeling, integrating user and course information to enhance prediction performance[15]. SahinM et al. developed an ANFIS model combining adaptive neural networks and fuzzy inference systems to predict learner dropout probability in MOOCs [16].Wen et al. used a feature matrix to store information related to learning behaviors, and employed a CNN to extract locally relevant features to improve the accuracy of MOOC dropout prediction [17]. Wang et al. proposed a dropout prediction model based on E-LSTM, which weights raw data by time intervals to more effectively fuse temporal information for dropout prediction[18]. Qian et al. developed a CLSA model based on the temporal data of learner behaviors for predicting dropout rates[19]. Mubarak et al. introduced a CONVLSTM model combining CNNs and LSTMs[20]. Xu et al. analyzed learners' behaviors over consecutive days, proposing the Lie group regional covariance matrix to represent local correlation information of learning behaviors and constructing a CNN model with a multi-coder pooling module to extract high-level features and improve prediction accuracy[21].

From the related research, it is evident that accurate analysis of student learning behavior in time series at a fine-grained level is crucial. Therefore, this study examines the relationship between different features and dropout issues, focusing on the time series prediction problem based on the temporal characteristics of learning in online teaching platforms.

III. Model Construction

In this study, deep learning techniques are employed to construct a network model for predicting online course dropout. Firstly, the convolutional neural network (CNN) is utilized to extract local features from the input data, leveraging its advantage in reducing the input data's output dimension, computational load, and training time. The data output from the CNN layer is then processed using the bidirectional long short-term memory network (BiLSTM), which excels at handling time series data. Finally, an attention mechanism is incorporated into the model to identify and allocate resources to the most important parts of the input information, thereby enhancing the prediction model's accuracy.

A. Input Layer

The primary task of the input layer is to process the feature data. The behavioral data of each student is organized in a weekly sequence for the course and defined as M. M represents a behavioral feature matrix.

$$M = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \quad (1)$$

Once the experimental data is integrated and the input format is defined, the data is divided, and the segmented training data is used as input for the CNN network layer. Each row of this data denoted as a_{mn}(where $n \in 1,2,\ldots,n$) represents the temporal feature data for n weeks.

B. Convolution and Pool Layer

These layers process the behavioral feature data of students, effectively extracting local features. The CNN network layer leverages its strengths in local feature extraction through convolutional and pooling layers. The ReLU function not only alleviates the problem of gradient vanishing but also accelerates the model's convergence speed, enhancing its nonlinear modeling capability[22]. The formula is shown below:

$$f(x) = max(0, x) = \begin{cases} 0, & x \leq 0 \\ x, & x > 0 \end{cases} \quad (2)$$

In the current network layer, the ReLU function is used as the activation function.

C. BiLSTM Network Layer

The BiLSTM network layer primarily trains the output feature data from the CNN layer to identify behavioral features that significantly impact the prediction of online course dropout. It deeply analyzes the bidirectional dependencies between student learning features in online learning.

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D. Attention Layer[23]

The attention mechanism is applied to process the feature vectors output from the BiLSTM layer. Through the calculation of attention scores, varying weights are assigned to input information, allowing the model to focus more on enhancing information pertinent to prediction outcomes, thereby improving prediction accuracy.

E. Output Layer

In the current network layer, the sigmoid function, commonly used in binary classification problems, is employed to output the probability value for each classification result. The calculation formula is presented below.

$$\text{sigmoid(x)} = \frac{1}{1+e^{-x}} \qquad (3)$$

The results of the Sigmoid calculation are transformed into corresponding dropout labels. If the probability value of the classification result exceeds 0.5, the student's dropout label is set to 1, indicating a higher risk. Conversely, if the probability value is below 0.5, the student's dropout label is designated as 0, indicating a lower risk.

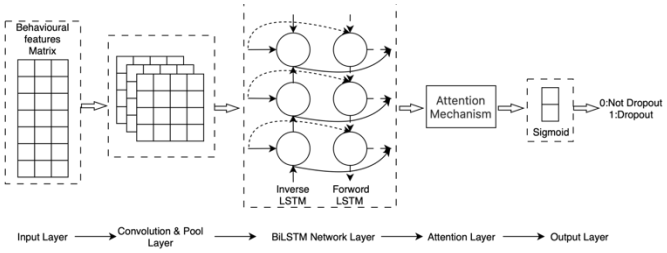


Fig. 1. overall architecture of CNN-BiLSTM-ATT model

The cross-entropy loss function is utilized to evaluate the accuracy of the dropout prediction model throughout the model training process.

IV. Experimental Results and Analysis

In this study, feature engineering is employed to construct essential behavioral data from online platforms. Subsequently, one-dimensional time series feature vectors are concatenated based on time units to build a prediction model utilizing deep learning algorithms, such as BiLSTM, renowned for their effectiveness in addressing time series prediction challenges.

A. Data preprocessing and temporal feature selection

Among the publicly available MOOC datasets, the dataset from the Schoolhouse Online MOOC platform (e.g., MOOCCube 2020) stands out for its richness in information and higher quality[24]. Numerous researchers have conducted dropout prediction studies based on this dataset. This study focuses on data preprocessing and feature selection to enhance the quality of experimental data at the data level. The experimental data were drawn from 10 courses within the dataset, selected for their abundance of video resources and higher student engagement. Using the data analysis software Anaconda and the Pandas toolkit, features were parsed into weekly features. Each week comprises 10 time-series features representing student learning behavior. These features include the frequency of course access, duration of course access, engagement with other modules, and engagement with assignments. The specific fields are detailed in Table 1.

TABLE I. Data Set Temporal Feature Selection

Feature number	Name	Feature description
S1	Number of times per week accessing the course	Number of times accessing the course during the learning process
S2	Hours of access to the course per week	Hours of access to the course during the MOOC learning process
S3	Number of visits to other modules of the course per week	Number of visits to other modules of the course during the learning process
S4	Number of times per week accessing assignments	Number of times assignments are accessed during the MOOC learning process
S5	Number of visits to forums per week	Number of visits to forums corresponding to the course during the MOOC study program
S6	Participation in discussions per week	Participation in discussions in the online platform
S7	Number of active days per week	Number of active days in the MOOC learning process
S8	Number of webpage closures per week	Number of webpage closures during the MOOC study program
S9	Number of times per week watching videos	Number of times per week watching instructional videos during the MOOC learning process
S10	Time spent watching videos per week	Time spent watching videos

Through preprocessing of the data, the quality of the experimental data can be enhanced at the data level. Utilizing the aforementioned temporal features, the experiment segments the feature data on a weekly basis, resulting in the creation of a one-dimensional temporal feature vector with a length of 10k+2, as illustrated in Figure 2.

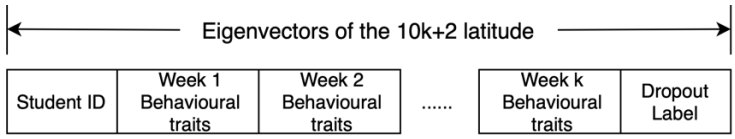


Fig. 2. Time series representation

The feature vector comprises three primary components:

1) Student Number: This serves as a unique identifier for each time-series feature vector.

2) Features: This section records the temporal feature data for each student over k weeks, resulting in a total data length of 10k, where k represents the number of weeks.

3) Dropout Labe: This label is utilized by the predictive model to denote a student's dropout status in the current course, with 0 indicating no dropout and 1 indicating dropout.

These data encompass 1,463,242 records from 48,326 users. The dataset was divided into a training set and a test set in a 7:3 ratio.

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B. Evaluation metrics

To evaluate the generalization performance of the prediction model in this study, reliance on evaluation metrics measuring its generalization ability is essential. Commonly employed for assessing binary classification performance, this study adopts Accuracy, Precision, and Recall as evaluation metrics[25]. Below are the relevant formulas:

$$Accuracy = \frac{TP+TN}{TP+FP+TN+FN} \quad (4)$$

$$Precision = \frac{TP}{TP+FP} \quad (5)$$

$$Recall = \frac{TP}{TP+FN} \quad (6)$$

To account for the balance between the two metrics, the F1 (F1-Score) value is introduced as an evaluation metric. This metric is calculated from the precision and recall rates using the following formula:

$$F1 = \frac{2\times Precision\times Recall}{\text{Precision+Recall}} \quad (7)$$

The results of the two dropout predictions, as depicted in the above formulas for TP, FP, TN, and FN, are represented by a confusion matrix as illustrated below.

TABLE II. CONFUSION MATRIX

	Predicted Dropout(1)	Not Dropout(0)
Actual Dropout(1)	TP	FN
Not Dropout(0)	FP	TN

C. experimental environment

The data and code deployment for this experiment are conducted on a local computer with the following hardware specifications: an Apple M1 Max CPU and a 500GB hard disk. The software environment includes Python 3.9, and the code is written in Jupyter Notebook. The deep learning framework used is PyTorch.

D. Experimental setup

Ten MOOC courses were selected for the experiment to evaluate the effectiveness of the CNN-BiLSTM-ATT model in predicting MOOC dropouts. The experiment inputs the time-series features of each MOOC course into the model. These time-series features are then combined into a one-dimensional time-series feature vector in chronological order and normalized to eliminate differences caused by varying scales, thereby improving the convergence speed of the prediction model. Given the sparsity of the data, 70% of the student behavior data in the dataset is used as training data, while the remaining 30% is used for testing to obtain the final prediction results for MOOC dropouts.

This study evaluates the effectiveness of the proposed CNN-BiLSTM-ATT prediction model by comparing its prediction results with those of four benchmark models: logistic regression (LR), support vector machine (SVM), decision tree model, and convolutional neural network (CNN)[26][27].The following table lists the three hyperparameters used during the experiment.

TABLE III. MODEL HYPERPARAMETER SETTINGS

Parameter	value
batch_size	80
epoch	EarlyStopping
dropout	0.3

E. Results and analysis

The experiments in this study use the time-series data constructed from the relevant courses as inputs to validate the prediction effectiveness of the CNN-BiLSTM-ATT model. Three experiments were conducted with the model, and the average of these three experiments was taken as the final experimental result.

TABLE IV. COMPARISON OF EXPERIMENTAL RESULTS

Model	Accuracy	Precision	Recall	F1-score
logistic Regression（LR）	0.681	0.676	0.681	0.681
Support Vector Machine（SVM）	0.691	0.672	0.690	0.690
Decision Tree(DT）	0.632	0.641	0.632	0.632
convolutional neural networks(CNN)	0.864	0.852	0.864	0.864
CNN-LSTM	0.871	0.864	0.871	0.867
CNN-BiLSTM-ATT	0.887	0.878	0.887	0.887

As shown in Table 4, the overall performance of the CNN-BiLSTM-ATT model surpasses that of the other models. The CNN-BiLSTM-ATT model outperforms the five benchmark models in all evaluation metrics.

- The overall performance of the CNN-BiLSTM-ATT model is approximately 20% higher compared to the LR, SVM, and DT models, indicating a significant advantage over traditional machine learning models in classifying dropout problems for online course learning.
- Compared to the CNN model, the CNN-BiLSTM-ATT model improved accuracy, precision, recall, and F1-Score by 1.3%, 1.4%, 1.7%, and 1.7%, respectively.
- Compared to the CNN-LSTM model, the CNN-LSTM-ATT model incorporates an attention mechanism, which assigns varying weights to different parts of the input data. This allows the model to prioritize information crucial for prediction, enhancing the connection between the model and the original data, and improving the F1 score by approximately 1%.

These results demonstrate that the CNN-BiLSTM-ATT model significantly enhances the classification performance of dropout problems in online course learning after applying feature engineering and the designed 2D matrix inputs. Thus, the

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CNN-BiLSTM-ATT model can more effectively predict dropout issues in online course learning.

CONCLUSION

This paper explores the application of deep learning algorithms to the problem of MOOC (Massive Open Online Course) dropout prediction, particularly from a time series prediction perspective. The study initially converts student behavioral data into time series data through feature engineering. Subsequently, a comprehensive CNN-BiLSTM-ATT dropout prediction model is constructed by leveraging the local feature extraction capability of convolutional neural networks (CNN), the bidirectional long short-term memory network's (BiLSTM) advantage in processing time series data, and the Attention Mechanism (ATT). Experimental results indicate that the CNN-BiLSTM-ATT model achieves an average accuracy of 88.7% in MOOC dropout prediction. Compared to traditional machine learning models, such as LR, SVM, and DT, the CNN-BiLSTM-ATT model significantly improves both accuracy and F1 score. Additionally, it outperforms typical deep learning algorithms, namely CNN alone and CNN-LSTM models, in terms of accuracy and F1 score. These results demonstrate the superior performance of the CNN-BiLSTM-ATT model in predicting student dropout behavior.

In addition to this, this study can be explored in depth in the future in the following two areas:

1) Correlation Studies of Learning Behaviors with Achievement: Current research has primarily focused on analyzing learner dropout due to limitations in available datasets. Future studies could extend to correlating learning behavior data with learner achievement. This would include not only predicting learner performance but also exploring potential links between academic performance and learning behaviors. Such research aims to identify key factors affecting learning effectiveness and provide data to support the improvement of teaching methods.

2) Development of Personalized Teaching Recommendation Systems: Although the models and methods studied in this paper have not yet been applied to actual MOOC platforms, future work will involve in-depth analyses of the learning characteristics of different student groups. Based on these characteristics, researchers can develop personalized teaching recommendation systems tailored to the learning needs of various students, thereby enhancing the teaching effectiveness and student satisfaction of MOOC platforms.

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